

Kai Zhu

PH. D. STUDENT · ZHEJIANG UNIVERSITY

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“Unveiling nature through atomistic simulation and machine learning.”

Summary

I am a Ph.D. student in Pharmaceutics at Zhejiang University, advised by Prof. Hou Tingjun. I am also a visiting student at the Italian Institute of Technology, working with Luigi Bonati and Prof. Michele Parrinello. My research focuses on developing machine learning-based enhanced sampling methods and their applications to atomic systems.

Education

Atomistic Simulations Group, Italian Institute of Technology

JOINT PH.D. STUDENT

Genova, Italy

Dec. 2025 – Dec. 2027

College of Pharmaceutical Sciences, Zhejiang University

Hangzhou, China

PH.D. IN PHARMACEUTICS

Sep. 2023 – Jun. 2028

College of Pharmaceutical Sciences, Wenzhou Medical University

Wenzhou, China

B.S. IN PHARMACEUTICS

Sep. 2019 – Jun. 2023

Research Experience

Italian Institute of Technology

Genova, Italy

RESEARCH INTERN, ADVISED BY PHD. LUIGI BONATI AND PROF. MICHELE PARRINELLO

Dec. 2025 – Dec. 2027

Research Topics: machine learning, enhanced sampling, atomic simulation

Zhejiang University

Hangzhou, China

PHD STUDENT, ADVISED BY PROF. TINGJUN HOU AND PROF. DAN LI

Jun. 2023 – Jun. 2025

Research Topics: machine learning, enhanced sampling, intrinsically disordered proteins

Honors & Awards

DOMESTIC

2025 Qiu Shi Flying Eagle Scholarship, Zhejiang University

Hangzhou, China

Publication

† Co-first author; * Corresponding author;

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| 2026 | MetalloDock: Decoding Metalloprotein–Ligand Interactions via Physics-Aware Deep Learning for Metalloprotein Drug Discovery
H. Zhang†, Xujun Zhang†, Qun Su†, Yangyang Zheng, Linlong Jiang, K. Zhu , et al. Journal of the American Chemical Society | [6] |
| 2025 | Enhanced Sampling in the Age of Machine Learning: Algorithms and Applications
K. Zhu† , E. Trizio†, J. Zhang, R. Hu, L. Jiang, T. Hou*, L. Bonati*. Chemical Reviews | [5] |
| 2025 | A Scalable and Quantum-Accurate Foundation Model for Biomolecular Force Field via Linearly Tensorized Quadrangle Attention
Q. Su†, K. Zhu† , Q. Gou†, ..., J. Wang*, C. Hsieh*, T. Hou*. ARXIV PREPRINT 2507.00884. | [4] |
| 2025 | Self-Supervised Evolution Operator Learning for High-Dimensional Dynamical Systems
G. Turri, L. Bonati, K. Zhu , M. Pontil, P. Novelli* ARXIV PREPRINT 2505.1867. | [3] |

- 2025 **Revisiting Protein–Protein Docking: A Systematic Evaluation Framework.** [2]
L. Jiang†, K. Zhang†, **K. Zhu**, Y. Wang, Y. Kang*, T. Hou*. **Journal of Chemical Information and Modeling** 65(19), 10573–10587.
- 2025 **From Traditional Methods to Deep Learning Approaches: Advances in Protein–Protein Docking.** [1]
L. Jiang, K. Zhang, **K. Zhu**, H. Zhang, C. Shen*, Tingjun Hou*. **Wiley Interdisciplinary Reviews: Computational Molecular Science** 15(2), e70016.

Skills

Programming	Python, C++, C, $\LaTeX$, Linux
Frameworks	PyTorch, TensorFlow, NumPy, Git, Anaconda
Atomic simulation	GROMACS, PLUMED, AMBER, OpenMM, ASE, LAMMPS, CP2K